
The Benchmark Lottery

Mostafa Dehghani*, Yi Tay*, Alexey A. Gritsenko*, Zhe Zhao, Neil Houlsby,
Fernando Diaz, Donald Metzler†, Oriol Vinyals†
Google Research & DeepMind
{dehghani, yitay, agritsenko}@google.com

Abstract

The world of empirical machine learning (ML) strongly relies on benchmarks in order to determine the relative effectiveness of different algorithms and methods. This paper proposes the notion of a *benchmark lottery* that describes the overall fragility of the ML benchmarking process. The benchmark lottery postulates that many factors, other than fundamental algorithmic superiority, may lead to a method being perceived as better. On multiple benchmark setups that are prevalent in the ML community, we show that the relative performance of algorithms may be altered significantly simply by choosing different benchmark tasks, highlighting the fragility of the current paradigms and potential fallacious interpretation derived from benchmarking ML methods. Given that every benchmark makes a statement about what it perceives to be important, we argue that this might lead to biased progress in the community. We discuss implications of the observed phenomena and provide recommendations on mitigating them using multiple machine learning domains and communities as use cases, including natural language processing, computer vision, information retrieval, recommender systems and reinforcement learning.

1 Introduction

Quantitative evaluation is a cornerstone of machine learning research. As a result, benchmarks, including those based on data sets and simulations, have become fundamental to tracking the progress of machine learning research. Benchmarks have a long history in artificial intelligence research generally. There have been several attempts at designing milestones to capture progress toward artificial intelligence (e.g., human level game performance, the Turing test [Turing, 1950]). Specific system properties are measured through specialized benchmarks (e.g. for vision, natural language processing, robotics). All of these benchmarks, by design, encode values about what is salient and important, both across domains (e.g. natural language processing benchmarks versus robotics benchmarks) and within them (e.g. which languages are considered in an NLP benchmark, which environments are considered in a robotics benchmark).

As benchmarks become widely accepted, researchers adopt them, often without questioning their assumptions and algorithmic development becomes slowly tied to these success metrics. Indeed, over time, the research community makes collective decisions about what shared tasks—and values—are important (through peer review norms and resource investment) and which are not. Because of this, it is important for the research community to understand the individual, community, social, and political pressures that **influence why some benchmarks become canonical and others do not**. This paper argues that a meta-level understanding of benchmarks is a prerequisite for understanding of how progress in machine learning is made.

This paper presents a meta-analysis on how benchmarks may affect the direction and pace of progress in machine learning and puts forward the notion of a benchmark lottery. We argue that many factors other than the algorithmic superiority of a method may influence the emergence of algorithms that are

*Equal contribution, †equal advising.

perceived as better. Moreover, we claim that for a method to emerge successful, it has to first win the *benchmark lottery*. Out of the many potential trials in this lottery, a method has to be first well-aligned with the suite of benchmarks that the community has accepted as canonical. We refer to the alignment between the tasks brought forth by the community and successful algorithms as the *task selection* bias. We empirically show that the task selection process has a great influence over the relative performance of different methods. Moreover, we argue that benchmarks are *stateful*, meaning that the method has to also participate in the lottery at the right moment, and to align well with existing techniques, tricks and state-of-the-art. Related to this, we also briefly discuss how benchmark reuse may affect the statistical validity of the results of new methods.

On a whole, as we researchers continue to participate in the benchmark lottery, there are long-term implications, which we believe are important to be explicitly aware of. As such, the main goals of this paper are to (i) raise awareness of these phenomena and potential issues they create; and to, (ii) provide some recommendations for mitigating these issues. We argue that community forces and task selection biases, if left unchecked, may lead to unwarranted overemphasis of certain types of models and to unfairly hinder the growth of other classes of models - which may be important for making fast and reliable progress in machine learning.

The notion of what makes a benchmark canonical, in the sense that is widely accepted by the community, is also diverse depending on the field of study. On one hand, fields like natural language processing (NLP) or computer vision (CV) have well-established benchmarks for certain problems. On the other hand, fields such as recommender systems or reinforcement learning tend to allow researchers more freedom in choosing their own tasks and evaluation criteria for comparing methods. We show how this may act as *rigging the lottery*, where researchers can “make their own luck” by fitting benchmarks and experimental setups to models instead.

Overall, this paper explores these aspects of model evaluation in machine learning research. We frame this from a new perspective of the *benchmark lottery*. While there have been recent work that peers deeply into the benchmark tasks themselves [Bowman and Dahl, 2021], this work takes meta- and macro-perspectives to encompass factors that go beyond designing reliable standalone tasks. The remainder of the paper is organized as follow:

- Section 2 discusses how benchmarks can influence long-term research directions in a given (sub-)field, and describes the life cycle of a benchmark.
- Section 3 introduces the *task selection bias* and using established benchmark as examples shows how relative performance of algorithms is affected by the task selection process.
- Section 4 takes another view of the task selection bias and proposes *community bias* as a higher-level process that influences task selection. We show that forces from the broader research community directly impact the task selection process and as a result, play a substantial role in the creating the lottery.
- Section 5 posits that benchmarks are stateful entities and that participation in a benchmark differs vastly depending upon its state. We additionally discuss that continual re-use of the same benchmark may be problematic.
- Section 6 discusses *rigging the lottery*, the issue that some communities (e.g. recommender systems and reinforcement learning) face, where the lack of a well-established community-driven sets of benchmarks or clear guidelines may inadvertently enable researchers to fit benchmarks to model. We highlight the potential drawbacks of such an approach.
- Finally, in Section 7 we provide recommendations for finding a way out of the lottery by building better benchmarks and rendering more accurate judgments when comparing models.

Overall, unified benchmarks have led to incredible progress and breakthroughs in machine learning and artificial intelligence research [Devlin et al., 2018, Raffel et al., 2019, Brown et al., 2020, Dosovitskiy et al., 2020]. There is certainly a lot of benefits of having the community come together to solve shared tasks and benchmarks. Given that the role of benchmarks are indispensable and highly important for measuring progress, this work seeks to examine, introspect and find ways to improve.

2 Background

Measuring progress is one of the most difficult aspects of empirical computer science and machine learning. Such questions as “What are the best setup and task to use for evaluation?” [Ponce et al., 2006, Machado et al., 2018, Lin, 2019, Bowman and Dahl, 2021, Recht et al., 2019, Lin et al., 2021b, Gulcehre et al., 2020, Perazzi et al., 2016, Vania et al., 2020], “Which data or benchmark are most applicable?” [Metzler and Kurland, 2012, Beyer et al., 2020, Northcutt et al., 2021, Gulcehre et al.,

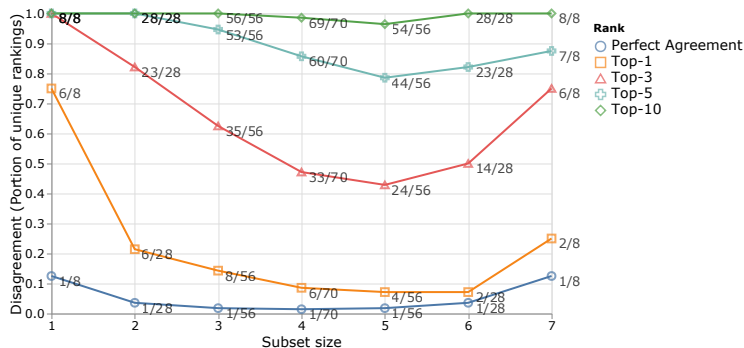


Figure 1: Disagreement of model rankings on the SuperGLUE benchmark as a function of the number of selected benchmark tasks. The x -axis represents the number of tasks in each sub-selection of tasks and each line corresponds to a different value of k for the Top- k in the rankings. Points are labels as A/B , where A is the number of unique model rankings and B is the total number of possible task combinations for this subset size. If $A = 1$, then all rankings are equivalent and consistent across all task selections; higher values of A correspond to higher degrees of disagreement between models rankings.

2020], “Which metrics are suitable?” [Machado et al., 2018, Bouthillier et al., 2021, Balduzzi et al., 2018, Bouthillier et al., 2019], or “What are the best practices for fair banchmarking?” [Torrallba and Efros, 2011, Armstrong et al., 2009b, Machado et al., 2018, Sculley et al., 2018, Lin, 2019, Bowman and Dahl, 2021, Bouthillier et al., 2021, Recht et al., 2019, Lin et al., 2021b, Balduzzi et al., 2018, Lipton and Steinhart, 2018, Bouthillier et al., 2019, Vania et al., 2020] are of utmost importance to correct empirical evaluation of new ideas and algorithms, and have been extensively studied. Nevertheless, the jury is still out on most of these questions.

We argue that some models and algorithms are not inherently superior to their alternatives, but are instead perceived as such by the research community due to various factors that we discuss in this paper. One of these factors is the software and hardware support for an idea, as captured in the concept of hardware lottery by Hooker [2020]. Here however we focus mainly on *benchmarking*-related factors, and discuss the role they play in the selection of a model as “fashionable” in the research world, and how this is often conflated with the model being better. When a class of models or algorithms gets recognition in the community, there will be more follow up research, adaption to more setups, more tuning and discovery of better configurations, which lead to better results. This is a valid way of propelling the field further. However, a question that we should also ask is how much progress could have been made by investing the same amount of time, effort, computational resources and talent in a different class of models. In other words, assuming model development as a complex high-dimensional optimization process, in which researchers are exploring a fitness surface, the initial point as well as the fitness function are the key factors for ending up with better optima, and both these factors are highly affected by the benchmarks used for evaluation. Next, we diving into various aspects of the benchmark lottery.

3 Task selection bias

As we show in this section, relative model performance is highly sensitive to the choice of tasks and datasets it is measured on. As a result, the selection of well-established benchmarks plays a more important role than is perhaps acknowledged, and constitutes a form of partiality and bias - the *task selection bias*.

3.1 SuperGLUE

In order to study the effect of aggregated scores and how findings change by emphasizing and de-emphasizing certain tasks, we explore the SuperGLUE dataset [Wang et al., 2019]. To demonstrate the task selection bias on this benchmark, we re-compute the aggregated scores using different combination of eight SuperGLUE tasks.

Model Runs. We consider over 55 different Transformer variations and modifications described in [Narang et al., 2021] and consider the fine-grained scores of each individual task on SuperGLUE. The analyzed data consists of the final performance of each of the 55 models on the 8 tasks of SuperGLUE. For each combination of tasks, we take a mean-aggregate model performance for all models on the selected tasks and produce a ranking of all 55 models. To make this ranking more meaningful, we only consider its Top- k entries, where $k \in \{1, 3, 5, 10\}$.

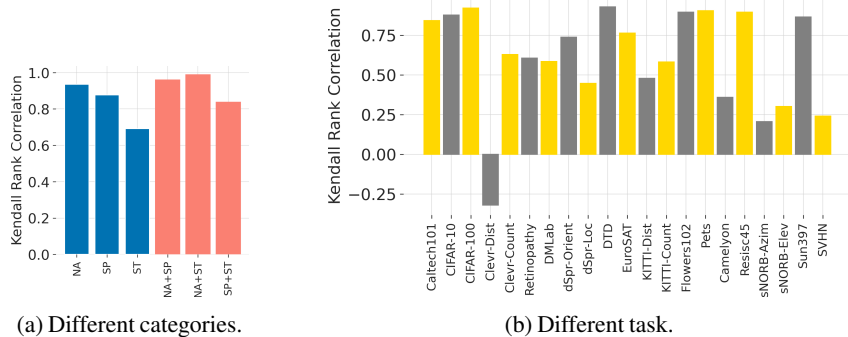


Figure 2: Rank correlation between the full VTAB score and the score for subsets of the benchmark.

Ranking inconsistency. Figure 1 gives a concise overview of the number of unique Top- k rankings produced obtained from fixed-size subsets of tasks. For example among the 70 different possibilities of selecting 4 out of 8 tasks, there are 6 distinct model ranking orders produced for Top-1 (i.e. there are 6 different possible top models). Moreover, when considering Top-3 or even Top-5, almost 60 out of 70 rankings do not agree with each other. Overall, the rankings become highly diverse as the subset of task selected from the benchmark is varied. This forms the core of the empirical evidence of the task selection bias.

3.2 Visual Task Adaptation Benchmark (VTAB)

A similar situation can be observed for the Visual Task Adaptation Benchmark (VTAB; Zhai et al. [2019]) benchmark. VTAB is used for evaluating the quality of representations learned by different models in terms of their ability to adapt to diverse, unseen tasks with few examples. VTAB defines a total of 19 tasks, grouped into three categories: *Natural*, *Specialized*, and *Structured*. We have evaluated 32 different models against all the 19 VTAB tasks. More details on the tasks, categories, and models can be found in Appendix A.

First, we study the agreement of the integrated score across all 19 tasks with the integrated scores obtained from different combinations of the three task categories: natural (NA), specialized (SP), and structured (ST). Figure 2a shows the Kendall rank correlation, when ranking different models based on the full VTAB score and based on the category (combination) score. It can be seen that rankings of models based on different combinations of categories are not always perfectly correlated. For instance, the structured (ST) subcategory has a correlation of ≈ 0.7 with the full VTAB score, thus highlighting rather different aspects of the competing models. But a more striking point here is the wide variance in correlation between different tasks. This shows that crowning a model as the winner based on the single score can be suboptimal, and demonstrates how the random nature of task selection can become a lottery that algorithms need to win. More analyses on the agreement of different combination of subcategories on the top- k models can be found in Appendix A.4.

Figure 2b also presents the correlations between the rankings based on the individual tasks and the integrated VTAB score. Unsurprisingly, an even stronger disagreement between rankings is observed (mean Kendall correlation of ≈ 0.60), including tasks with negative correlation.

One of the main difficulties in integrating scores of multiple tasks is the incorporation of individual task difficulty into the integrated score. This is made more complex by the fact that there are multiple facets to what makes a task difficult. For instance, in the case VTAB, distribution shift between the pretraining dataset and the downstream dataset, e.g. structured and specialized tasks are further outside the training distribution when the model is pretrained on the ImageNet-21k dataset. Alternatively, the size of the training data for downstream is another aspect influencing the task difficulty. An even simpler and commonplace aspect of task difficulty is the number of prediction classes in this dataset, and consequently the score for a random baseline for the task. In the case of VTAB the authors should be credited for investigating some of the failure modes of using a single mean to aggregate for the entire benchmark, such as the fact that some tasks use the same input data (e.g. CLEVR-Count and CLEVR-Dist), thus upweighting those domains; that the task groups differ in size; or that the tasks exhibit different performance ranges across methods. Kolesnikov et al. [2019] tried different ranking aggregation methods from [Balduzzi et al., 2018] and showed they were fairly correlated.

For more analyses and additional case studies (Long Range Arena and RL-Unplugged) check Appendix A.

4 Community bias

Even when viewed as a random process, the task selection bias described in Section 3 alone is sufficient for creating arbitrary selection pressures for machine learning models. We argue however that there is also a higher-level process in which the broader research community influences the task selection, and that counter intuitively **leads to the lottery forces not being diminished, but instead more pronounced**. This section takes a people perspective of the benchmark lottery and postulates that it is not only the “gamemasters” (benchmark proposers) but also the community that contribute to and reinforce it.

While researchers technically have the freedom to select any dataset to showcase their method, this choice is often moderated by the community. A common feedback in the review process of scientific publications that any ML researcher will face eventually is a criticism on the choice of benchmark. For example “*the method was not evaluated on X or Y dataset*” or “*the method’s performance is not SOTA on dataset Z*”. Over time, ML researchers tend to gravitate to safe choices of tasks and benchmarks. For example, most papers proposing new pretrained language models [Lan et al., 2019, Liu et al., 2019, Clark et al., 2020, Yang et al., 2020] evaluate on GLUE even if alternatives exist (See example below for further substantiation). To this end, the community might ‘double down’ on a benchmark where it becomes almost an unspoken rule for one to evaluate on a particular benchmark. **Once a benchmark builds up a following and becomes well-established**, it is possible that reviewers will continue to ask for results on these benchmarks, potentially regardless of suitability and/or appropriateness. This makes it difficult to fix potentially broken benchmarks.

As foreshadowed, commonly used benchmarks are not immune to containing errors. While these errors are likely to be small (as otherwise they would presumably be noticed early on), they do matter in close calls between competing methods. Northcutt et al. [2021] identified label errors in test sets of 10 of the most commonly-used computer vision, natural language, and audio datasets; for example, there are label errors in 6% of the examples in the ImageNet validation set. They showed that correcting label errors in these benchmarks changes model ranking, especially for models that had similar performance. **In the field of NLP, it was later found in SNLI [Bowman et al., 2015] a large amount of annotation artifacts exists, and it is possible to simply infer the correct label by only using the premise and not the hypothesis [Gururangan et al., 2018]**. It is worth noting that SNLI, being the canonical benchmark for natural language inference (NLI), was easily perceived as mandatory for almost any research on NLI.

Naturally, this problem is not only restricted to the peer review process, but also extends to the public perception of papers after they are published regardless of whether they went through peer review process or not. The community bias problem here simply means that the community collectively assigns a weighted impact score for doing well on arbitrarily selected tasks. Achieving state of the art on task Y is then deemed significantly less meaningful than doing that for task X . Moreover this is done without any explicit reasoning as to why one task is preferred to the other, or how even how such “decision” was made.

The main concern with respect to the community bias is that research is becoming too incremental and biased toward the common expectations, since a completely new approach will initially have a hard time competing against established and carefully fine-tuned models. For more discussion and concrete examples on the community bias check Appendix B.

5 Benchmarks are stateful

With leaderboards and continuous publication of new methods, it is clear that benchmarks are stateful entities as a model attempt at beating a benchmark at any given time depend on the information gathered from previous submissions and publications. This is a natural way of making progress on a given problem. For many machine learning benchmarks, researchers have full access to the holdout set. **Although not explicitly, this typically leads to the violation of the most basic datum of “one should not train on test/holdout set” by getting inspiration from already published works by others who presumably report only the best of the numerous models they evaluated on the test set.**

Beyond that, it is common to copy-paste hyper-parameters, use the same code, and more recently to even start from **pre-retrained checkpoints of previous successful models**. In such setups, where the discovery of new models is built on top of thousands of queries, direct or indirect, to the test set, the error rate on test data does not necessarily reflect the true population error [Arora and Zhang, 2021, Blum and Hardt, 2015, Dwork et al., 2015]. The adaptive data analysis framework [Dwork et al., 2015] provides evaluation mechanisms with guaranteed upper bounds on the difference between average error on the test examples and the expected error on the full distribution (population error rates). **Based**

on this framework, if the test set has size N , and the designer of a new model can see the error of the first $i - 1$ models on the test set before designing the i -th model, one can ensure the accuracy of the i -th model on the test set is as high as $\Omega(\sqrt{i/N})$ by using the boosting attack [Blum and Hardt, 2015]. In other words, Dwork et al. [2015] state that once we have $i \gg N$ the results on the test set are no longer an indication of model quality. It has been argued that what matters is not only the number of times that a test set has been accessed as stated by adaptive data analysis, but also how it is accessed. Some empirical studies on some popular datasets [Recht et al., 2018, Yadav and Bottou, 2019, Recht et al., 2019] demonstrated that overfitting to holdout data is less of a concern than reasoning from what have suggested in [Blum and Hardt, 2015]. Roelofs et al. [2019] also studied the holdout reuse by analyzing data from machine learning competitions on the Kaggle and show no significant adaptive overfitting on the classification competitions. Other studies showed that additional factors may prevent adaptive overfitting to happen in practice. For instance, [Feldman et al., 2019b,a] show that in multi-class classification, the large number of classes makes it substantially harder to overfit due to test set reuse. In a recent study, Arora and Zhang [2021] argue that empirical studies that are based on creating or using new test sets (e.g. [Recht et al., 2018, Yadav and Bottou, 2019, Recht et al., 2019]), although reassuring in some level, are not always possible especially in datasets concerning rare or one-time phenomena. They emphasize the need for computing an effective upper bound for the difference between the test and population errors. They propose an upper bound using the description length of models that is based on the knowledge available to model designers before and after the creation of a test set.

All these studies considered, we can conclude that prior information about a model’s performance on the test set may influence future modeling choices and hyperparameter settings, and this adaptive behavior, in principle, may have an effect on generalization and can, to different extent, create a gap between test error and full distribution error.

From the benchmark lottery point of view, the most important aspect of the above phenomena is that the development of new models is shaped by the knowledge of the test errors of all models before it. There had been events in the past that accessing the test set more than others, intentionally, secured a margin for victory in the race². In other words, having the ability to access the test set more than others can be interpreted as buying more lottery tickets. Another aspect of benchmarks being stateful is that participating in shared tasks at a later stage is vastly different from the time of its inception. By then the landscape of research with respect to the specific benchmark is filled with tricks, complicated and specialized strategies, and know-how for obtaining top performance on the task. As an example, when the community realized that pre-training on MNLI is necessary for obtaining strong performance on RTE and STS datasets [Liu et al., 2019, Clark et al., 2020], and this became common practice later on. Experience shows that it is not uncommon for benchmark tasks to accumulate lists of best practices and tricks that are dataset- and task-specific. Many of the adapted recipes for scoring high are non-universal. Whether a novel algorithm is able to make use of these tricks (or whether they are available at all) is again a form of lottery, in which models that cannot incorporate *any* of the earlier tricks are significantly disadvantaged.

6 Rigging the lottery: making your own luck

For some tasks and problems, there are already standard benchmarks and established setups that are followed by most of the community. However, for some others, inconsistencies in the employed benchmarks or reported metrics can be observed. This diversity of evaluation paradigms makes comparisons between publications extremely difficult. Alternatively, in some cases there is simply no standard benchmark or setup, either because the problem is still young, or because there has never been an effort to unify the evaluation. Sometimes this is due to high computational cost of proper evaluation, like when reporting variance over multiple random seeds is important [Bouthillier et al., 2019]. While in other instances, the root cause is of behavioral nature, where researchers prefer to showcase only what their method shines at - oftentimes to avoid negative reviews, unsuccessful experiments, although performed, are simply not reported. Here, we study two known examples of this issue, which we refer as *rigging the lottery*.

6.1 Recommender systems and benchmark inconsistencies

The goal of recommender systems (RecSys) research is build systems that understand users’ interests and behaviors towards a list of potential items to recommend [Zhang et al., 2019]. Unlike the fields of NLP or CV, there are no unified well-established evaluation setups for recommender systems that provides canonical ranked lists of model performance. While there has been a famous Netflix

²<https://image-net.org/challenges/LSVRC/announcement-June-2-2015>

prize challenge³, this dataset or task have not been extensively used in academic research or for benchmarking new models. Moreover, even popular datasets like MovieLens [Harper and Konstan, 2015] or Amazon Reviews [He and McAuley, 2016] generally do not have a canonical test split, metric or evaluation method. Therefore, it is still quite unclear about which modern RecSys method one should adopt, as model comparisons are currently all over the place.

Furthermore, RecSys evaluation is also very challenging for a number of reasons. (i) Different recommendation platforms tackle slightly different problems (e.g retrieval [Yi et al., 2019], ranking ([Pei et al., 2019]), or multitask learning ([Zhao et al., 2019])), that each requires their own evaluation setup. (ii) As is common for user interacting systems, user’s reaction towards different algorithms can be different. Constructing offline datasets of user behaviors from an existing system creates an off-policy evaluating challenge [Swaminathan et al., 2016]. (iii) A real-world recommendation system trains on billions of users and items, the scale of user-item interactions makes it extremely difficult to create a complete dataset containing all possible user-item interactions [He et al., 2016]. As a result, evaluation setups in many recommender system papers tend to be arbitrary.

There exists a small number of public datasets (see Appendix D), such as MovieLens [Harper and Konstan, 2015] or Amazon Product Review [He and McAuley, 2016] that are commonly used for evaluating recommender systems. However, even these datasets are tweaked differently in various publications, leading sometimes to contradictory results [Rendle et al., 2020, Zheng et al., 2019]. For example, some papers use Hit Ratio and NDCG as evaluation metrics [He et al., 2017], while others resort to using Recall@K [Zheng et al., 2019]. Interestingly, in this particular example, the same methods reverse their performance when a different metric is used. Holdout/test sets can also be created differently, with some papers, for example, using random split [Beutel et al., 2017] and others using an out-of-time split [Zhang et al., 2020].

While majority of this paper discusses cases where a standardized benchmark may lead to biased progress in the ML community, here we instead discuss the *exact opposite* - implications of having no consensus datasets or evaluation setups. Having no unified benchmark for the community to make progress on has numerous flaws. To name a few, (i) this hinders progress in the field, while possibly (ii) creating an illusion of progress. It is not surprising that under these circumstances researchers (potentially unknowingly) tend to find good experimental setups that fit their models instead of the other way around.

For a case study on inconsistencies of the evaluation setup in ALE benchmark check Appendix C.

7 What can we do?

We present suggestions for improving the benchmarking process in way that makes it less of a lottery.

7.1 Investing in making guidelines

We believe that the risks of “rigging the lottery” that is described in Section 6 can be minimized by standardizing the recipe for creating and using benchmarks.

Guidelines for creating benchmarks. Investing into shared guidelines for creating new benchmarks can be extremely beneficial to the long-term health of the research community. In our view, such guidelines should include the current best practices, aspects that require special attention; an should highlight potential concerns for issues that may emerge in the future when different models and algorithms are applied on it. Fortunately, there have been some efforts on providing guidelines and best practices for making new benchmarks. For example, Zhang [2021] discusses the need for how robotic warehouse picking benchmarks should be designed. Kiela et al. [2021] proposed a framework that sets clear standards for making new tasks and benchmarks. Denton et al. [2020] look at the dataset construction process with respect to the concerns along the ethical and political dimensions. Bender and Friedman [2018] also proposed using data statements for NLP datasets in order to provide more context on data, software, and results interpretation.

Guidelines for benchmark usage. Besides the necessity of making guidelines for “how to make new benchmark”, it is important to have clear guidelines for “how to use a benchmark”. This would be a great help with reducing the instances of rigging the lottery prevalent in some domains (Section 6). There are also several efforts targeting this goal. For instance, Albrecht et al. [2015], Machado et al. [2018] propose specific standards for the ALE benchmark (discussed in Section C.1). Ethayarajh

³<https://netflixprize.com/index.html>

and Jurafsky [2020] argue against ranking models merely based on their performance and propose to always report *model size*, *energy efficiency*, *inference latency*, and metrics indicating model *robustness* and *generalization to the out-of-distribution data*. Gebru et al. [2018] proposed that every dataset be accompanied by a datasheet that documents its motivation, composition, collection process, recommended uses, etc with the goal of increasing the transparency and accountability, mitigating unwanted biases in ML systems, facilitating greater reproducibility, and helping researchers and practitioners select more appropriate datasets for their chosen tasks.

Guidelines for conferences and reviewers. There have been attempts to ameliorate the problems related to the benchmark lottery, especially its community biases and the statefulness aspects (Sections 4 and 5). For example, NLP conferences have specially called out “*not being SOTA*” as an invalid basis for paper rejection⁴. We believe it is possible to leverage education through the review process in order to alleviate many negative aspects of benchmark lottery. As an example, we can make sure that in the review process, scores on a particular benchmark are not used for an immediate comparison with the top-ranking method on that benchmark, but rather as a sanity check for new models and simply an efficient way of comparing against multiple baselines. This way, fundamentally new approaches will have a chance to develop and mature instead of being forced to compete for top performance right away or get rejected if not succeeded in the early attempts.

7.2 Statistical significance testing

Presence of established benchmarks and metrics alone does not necessarily lead to steady improvement of research ideas; it should also be accompanied by rigorous procedures for comparing these ideas on the said benchmarks. For example, Armstrong et al. [2009a] discuss the importance of comparing improvements to the strongest available baselines, however, the question of how do we know that if a new model B is *significantly* better than its predecessor model A remains anything but solved 10 years later [Lin et al., 2021a].

Benchmark results as random samples. Machine learning models are usually trained on a training set and evaluated on the corresponding held out test set, where some performance metric m is computed. Because model training is subject to sources of uncontrolled variance, the resulting metric m should be viewed as a single sample from the distribution describing this models performance. Because of that, deciding which of the two models is better based on point estimates of their performances m_A and m_B may be unreliable due to chance alone. Instead distributions of these metrics $p(m_A)$ and $p(m_B)$ can be compared using statistical significance testing to determine whether the chance that model A is at least as good as model B is low, i.e. $p(A \leq B) < \alpha$ for some *a priori* chosen significance level α . Estimation of $p(A \leq B)$ forms for crux of statistical significance testing. It can be done either by using parametric tests that make assumptions on distributions $p(m_A)$ and $p(m_B)$ and thus often need fewer samples from these distributions, or by using non-parametric tests that rely on directly estimating the metric distributions and require more samples.

Popularity of standardised benchmarks and exponential growth in the amount of research that the ML community has experienced in recent years⁵ exacerbate the risk of inadvertently misguiding research through lax standards on declaring a model as an improvement on the SOTA. Indeed, if point estimates are used in place of statistical significance testing procedures, sampling $m'_A \sim p(m_A)$ and $m'_B \sim p(m_B)$ such that $m'_B > m'_A$ is only a matter of time, even if performance of the two models is not actually different. Note that this is *not* the same as the issue described in Section 5, but could instead be thought of as winning a lottery if you purchase enough lottery tickets.

Beyond a single train-test split. Collecting strong empirical evidence that model B significantly outperforms model A amounts to running multiple trials of hyper-parameter optimisation over sources of variation such as dataset splits, data ordering, data augmentation, stochastic regularisation (e.g. dropout) and random initialisation to understand the models’ variance, and is prohibitively expensive⁶. Bouthillier et al. [2021] showed that dataset split contributes the most to model variance compared to other sources of variation. However, providing multiple dataset splits to estimate this variance is not standard practice in benchmark design. Systematic variance underestimation may lead to a series of false positives (i.e. incorrectly declaring a model to be a significant improvement) that stall research progress, or worse - lead the research community astray by innovating on “improved overfitting” in

⁴<https://2020.emnlp.org/blog/2020-05-17-write-good-reviews>.

⁵<https://neuripsconf.medium.com/what-we-learned-from-neurips-2020-reviewing-process-e24549eea38f>

⁶Although Bouthillier et al. [2021] also propose a pragmatic alternative to the exhaustive study of all source of variation.

place of algorithmic improvements. Going forward, one way of addressing this limitation is to design benchmarks with *multiple* fixed dataset splits. As an added benefit, model performance reported across such standardised splits would also enable the application of a variety of statistical tests not only within the same study, but also across publications.

Benchmark design with statistical testing in mind. The choice of a suitable statistical testing procedure is non-trivial. It must consider the distribution of the metric m that is being compared, the assumption that can be safely made about the distribution (i.e. whether a parametric test is applicable or a non-parametric test should be used), the number of statistical tests performed (i.e. whether multiple testing correction be employed) and can also change as the understanding of the metric evolves [Demšar, 2006, Bouthillier et al., 2021, Lin et al., 2021a, Dror et al., 2017]. We therefore recommend that benchmark design is accompanied by the recommendation of the suitable statistical testing procedures.

Beyond a single dataset. Often we are interested in understanding whether model B is significantly better than model A *across a range of tasks*. This kind of comparisons are facilitated by benchmarks that span multiple datasets (e.g. VTAB or GLUE). Already the question of what it means to do better on a multi-task benchmark is non-trivial due to the task selection bias (see Section 3) - is it sufficient for model B to do better on average; or should it outperform model A on all tasks? It is not surprising that the statistical testing procedures for such benchmarks are also more nuanced - the answer to this question leads to different procedures. For this reason more elaborate procedures are required. For example, for the case when we are interested in seeing whether B outperforms A on average Demšar [2006] propose to ignore the variance on individual datasets and treat the model A and B 's performance across datasets as samples from two distributions that should be compared. Alternatively, for cases when we are interested in seeing whether B is better than A on all datasets Dror et al. [2017] propose to perform statistical testing on each of the dataset separately while performing multiple testing correction. Here again the “right” statistical testing procedure depends on the benchmark, its composition and the criteria for preferring one model over another; and we believe that the community would benefit if these questions were explicitly answered during benchmark design.

7.3 Rise of living benchmarks

Another major issue for many popular benchmarks is “creeping overfitting”, as algorithms over time become too adapted to the dataset, essentially memorizing all its idiosyncrasies, and losing the ability to generalize. This is essentially related to the statefulness of benchmarks discussed in Section 5. Besides that, measuring progress can be sometimes chasing a moving target since the meaning of progress might change as the research landscape evolves. This problem can be greatly alleviated by for instance changing the dataset that is used for evaluation regularly, as it is done by many annual competitions or reoccurring evaluation venues, like WMT⁷ or TREC⁸. Besides that, withholding the test set and limiting the number of times a method can query the test set for evaluation on it can also potentially reduce the effect of adaptive overfitting and benchmark reuse. In a more general term, an effective approach is to turn our benchmarks into “living entities”. If a benchmark constantly evolves, for instance, adds new examples, adds new tasks, deprecates older data, and fixes labeling mistakes, it is less prone to “tricks” and highly robust models would find themselves consistently doing well across versions of the benchmark. As examples of a benchmark with such a dynamic nature, GEM is a living benchmark for natural language generation [Gehrmann et al., 2021] or Dynabench [Kiela et al., 2021] proposes putting humans and models in the data collection loop where we continuously reevaluate the problem that we really care about.

Epilogue

Ubiquitous access to benchmarks and datasets has been responsible for much of the recent progress in machine learning. We are observing constant emergence of new benchmarks. And on the one hand, the development of benchmarks is perhaps a sign of continued progress, but on the other hand, there is a danger of getting stuck in a vicious cycle of investing in making benchmark that soon will be rejected due to their flaws. We are in the midst of a data revolution, and have an opportunity to make faster progress towards the grand goals of artificial intelligence, if we understand the pitfalls of the current state of benchmarking in machine learning. The “benchmark lottery” provides just one of the narratives of struggling against benchmark-induced model selection bias. But there are many reasons to be excited about the future - the community is continuously taking positive delta changes that contribute to fixing issues with measuring progress in the empirical machine learning.

⁷<http://statmt.org/>

⁸<https://trec.nist.gov/>

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Checklist

1. For all authors...
 - (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]
 - (b) Did you describe the limitations of your work? [Yes]
 - (c) Did you discuss any potential negative societal impacts of your work? [Yes]
 - (d) Have you read the ethics review guidelines and ensured that your paper conforms to them? [Yes]
2. If you are including theoretical results...
 - (a) Did you state the full set of assumptions of all theoretical results? [N/A]
 - (b) Did you include complete proofs of all theoretical results? [N/A]
3. If you ran experiments (e.g. for benchmarks)...
 - (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? [N/A]
 - (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? [N/A]
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? [N/A]
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? [N/A]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets...
 - (a) If your work uses existing assets, did you cite the creators? [Yes]
 - (b) Did you mention the license of the assets? [Yes]
 - (c) Did you include any new assets either in the supplemental material or as a URL? [No]
 - (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? [N/A]
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? [N/A]
5. If you used crowdsourcing or conducted research with human subjects...
 - (a) Did you include the full text of instructions given to participants and screenshots, if applicable? [N/A]
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals, if applicable? [N/A]
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? [N/A]

Appendix

A Tasks selection bias: additional case studies and more analysis

A.1 SuperGLUE: Accuracy of models on different combinations of tasks

Figure 3 shows the performance of different models on different combinations of all tasks. The very top row in the heatmap is the SuperGLUE score (average over all tasks) and models on the x-axis are sorted based on their performance on the “All” score. However, there is no strong pattern observable in the plot, like models with the highest “All” score also usually perform best on other combinations of tasks.

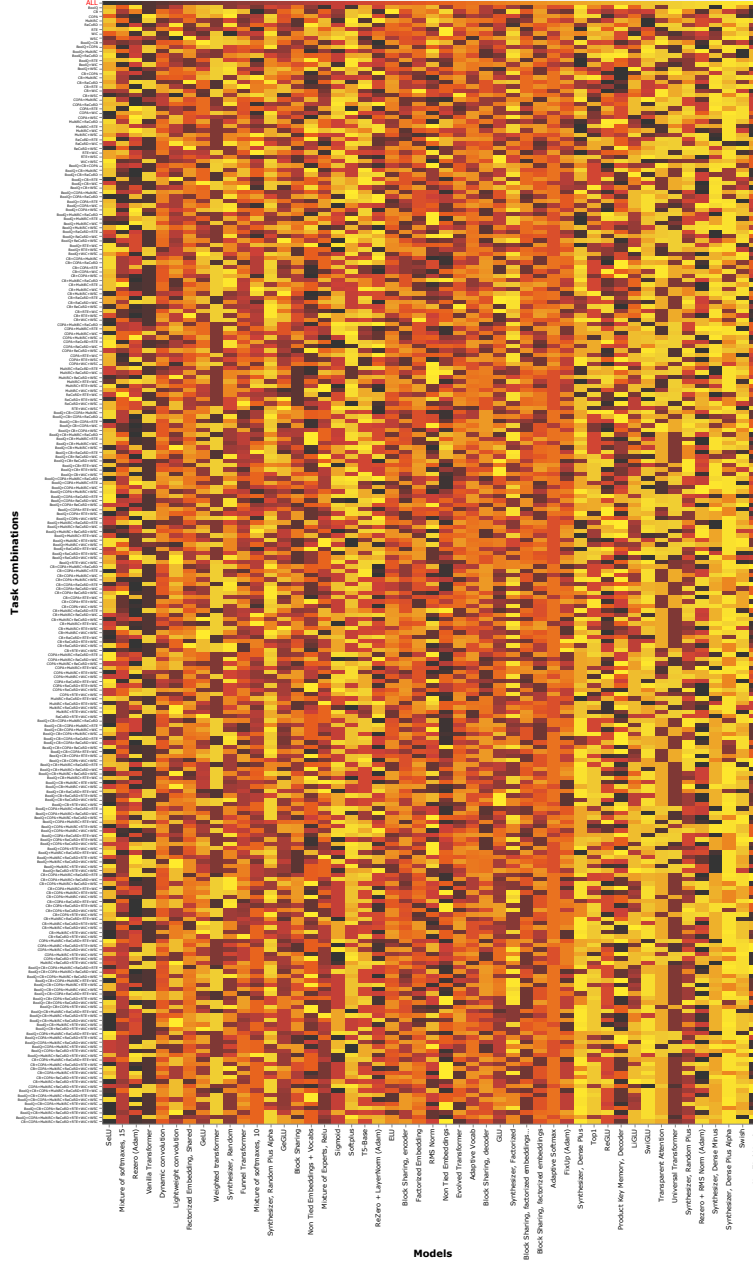


Figure 3: Performance of different models on different combinations of tasks in SuperGLUE benchmark. Models are sorted on the x -axis based on their performance in “ALL” tasks. We can observe that there no clear and strong pattern or correlation in different combinations of tasks compared to the full benchmark, indicating that there is no “best” model, while most of the time, the top-ranked model is simply taken as the winner.

A.2 SuperGLUE: Effect on relative ranking of models

A significant amount of work in machine learning modeling is to determine the relative performance of a set of different inductive biases or model architectures. We show that the ranking of models can be drastically altered based on the choice of the subset of the benchmark considered. In other words, the relative performance of models can be easily manipulated by task selection. In order to show this phenomenon, we select ten models, namely vanilla Transformers, Weighted Transformers, Funnel Transformers, Switch Transformers, Lightweight Convolutions, Dynamic Convolutions, Universal Transformers and Adaptive Softmax. The results are similarly obtained from [Narang et al., 2021]. Table 1 reports a sample of different selection of tasks. We show that for a different selection of tasks, the relative order of model performance is *very* different. Notably, models such as Universal, MoE, Adaptive Softmax, Switch all take turns to become the best performing model on multiple task configurations. Consequently, it is easy to see that with some manipulation of the benchmark configuration, it is easy to endorse and favor the performance of one model over another.

Table 1: Relative order of different models when selecting different subsets of SuperGLUE. Selecting different subsets of tasks can produce **very different** outcomes for relative ranking of model architectures. Models that did not appear in Top-5 at all are Lightweight Conv, Dynamic Conv and Transparent Attention. For tasks, A=BoolQ, B=CB, C=CoPA, D=MultiRC, E=ReCoRD, F=RTE, G=WiC, H=WSC.

Tasks	Top-5 Performing Models (In Order)
H	Universal, Switch, Adaptive Softmax, Weighted, Vanilla
G	MoE, Switch, Vanilla, Funnel, Universal
A, B	Adaptive Softmax, Vanilla, MoE, Switch, Weighted
A, C	MoE, Switch, Adaptive Softmax, Vanilla, Universal
D, H	Switch, Universal, Adaptive Softmax, MoE, Weighted
B, E, H	Adaptive Softmax, Switch, MoE, Vanilla, Weighted
F, G, H	Switch, MoE, Adaptive Softmax, Universal, Vanilla
A, F, G	MoE, Switch, Vanilla, Adaptive Softmax, Vanilla
C, F, G, H	Switch, MoE, Adaptive Softmax, Vanilla, Universal
A, C, D, G	MoE, Switch, Adaptive Softmax, Vanilla, Universal
All	Switch, MoE, Adaptive Softmax, Vanilla, Universal

Optimistically, we also note that even under the notion of a *lottery*, not all models have equal odds. Models that perform poorly across all tasks generally tend to not have a chance to qualify for the Top-5 of any of the above benchmark configurations. We note that Lightweight convolutions, Dynamic Convolutions, and Transparent Attention never made it to any of the Top-5 rankings. Models such as Funnel and Weighted also make very limited appearances. In short, we show empirically that benchmark suites can do pretty well in filtering model architectures that do poorly on most tasks.

A.3 VTAB: Details about the tasks, categories, and models.

VTAB is used for evaluating the quality of representations learned by different models in terms of their ability to adapt to diverse, unseen tasks with few examples. In addition to standard natural image tasks, like classification on ImageNet or CIFAR datasets, VTAB includes tasks that are related to sensorimotor control, medical imaging and scene understanding. The benchmark defines the score of algorithm as its expected performance over a known distribution of tasks that includes those that a human can solve, from visual input alone.

VTAB defines a total of 19 tasks, grouped into three categories: (i) *Natural*, which contains natural images captured using standard cameras that represent generic, fine-grained, or abstract objects [Caltech101 [Fei-Fei et al., 2006], CIFAR100 [Krizhevsky et al., 2009], DTD [Cimpoi et al., 2014], Flowers102 [Nilsback and Zisserman, 2008], Pets [Parkhi et al., 2012], Sun397 [Xiao et al., 2010], and SVHN [Netzer et al., 2011].]; (ii) *Specialized*, which contains images of the world that captured through specialist equipment [Remote sensing: Resisc45 [Cheng et al., 2017] and EuroSAT [Helber et al., 2019]: aerial images of the Earth captured using satellites or aerial photography; Medical: Patch Camelyon [Veeling et al., 2018], metastases detection from microscopy images, and Diabetic Retinopathy [Kaggle and EyePacs, 2015], retinopathy classification from fundus images.]; and finally (iii) *Structured*, which contains tasks that designed to assess comprehension of the structure of a scene, mostly generated syntactically using simulated environments [CLEVR [Johnson et al., 2017]: Simple

shapes rendered in a 3D scene, with two tasks: counting and depth prediction, *dSprites* [Higgins et al., 2016]: Simple black-and-white shapes rendered in 2D, with two tasks: location and orientation prediction, *SmallNORB* [LeCun et al., 2004]: Artificial objects viewed under varying conditions, with two tasks: object azimuth and camera-elevation prediction, *DMLab* [Beattie et al., 2016]: Frames from a rendered 3D maze. The task involves predicting the time for a pre-trained RL agent to navigate to an object, *KITTI* [Geiger et al., 2013]: frames captured from a car driver’s perspective and the task is to predict the depth of the nearest vehicle.]. We have evaluated 32 different models against all the 19 VTAB tasks. The difference between models is on their architectures (e.g. WAE-GAN [Tolstikhin et al., 2017] vs. VIVI [Tschannen et al., 2020]), their sizes (e.g. ResNet-50 vs. ResNet-101 [Kolesnikov et al., 2019]), or the dataset they were pre-trained on (e.g. ResNet-50 pretrained on ImageNet-21k vs. ResNet-50 pretrained on JFT [Kolesnikov et al., 2019]).

A.4 VTAB: Sub-categories agreement on top-ranked models

Similar to Figure 1, we looked into the disagreement of Top-1, 2, and 3 models based on different combinations of three VTAB sub-categories. The x -axis represents the number of sub-categories in each sub-selection and each line corresponds to a different value of k for the Top- k in the rankings. Points are labels as A/B , where A is the number of unique model rankings and B is the total number of possible sub-category combinations for this subset size (there are 3 combinations of size 1 and 3 combinations of size 2 categories).

In all cases, $1/3$ is the perfect agreement (there exists only 1 unique ranking). As it can be seen, the top-ranked model is consistent across all the combinations, however, when considering Top-3 models, the rankings diverge and there is a full disagreement between different combinations of sub-categories (i.e., 3 out of 3 rankings are unique).

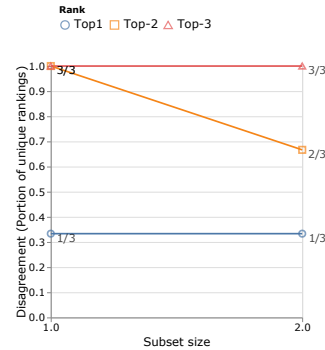


Figure 4: Disagreement of model rankings on the VTAB benchmark as a function of the number of selected benchmark sub-categories (Natural, Specialized, Structured).

A.5 Long Range Arena

The Long Range Arena (LRA; Tay et al. [2021a]) is a benchmark designed for aggregated evaluation of long-range Transformer models [Tay et al., 2020]. Similarly to other benchmark suites, LRA consists of six tasks: ListOps, Long Text Classification, Long Text Retrieval, Pixel-wise Image Classification and two variants of spatial reasoning based on the path-finder task. The authors rank eleven efficient transformer models by aggregating performance across all six tasks. To demonstrate that here too task selection matters, we computed Top-3 rankings of models for each task combination displayed in Table ???. Model name abbreviations are used for brevity and because the actual model names are not important for the purpose of this analysis. Notably, it is easy to see that the identity of each of the Top-3 changes frequently as the subset of evaluation tasks is changed.

A.6 RL Unplugged

RL Unplugged [Gulcehre et al., 2020] is a suite of benchmarks for offline reinforcement learning, where the task for the agent is to learn a policy directly from some logged data that is produced by a system as part of its normal operation, without interacting with the environment at the time of learning. RL Unplugged introduces a collection of task domains and associated datasets together with a clear evaluation protocol. It includes some widely used domains such as the DM Control Suite [Tassa et al., 2018] and Atari 2600 games [Bellemare et al., 2013], as well as Real-World RL (RWRL) tasks [Dulac-Arnold et al., 2019] and DM Locomotion tasks [Heess et al., 2017].

Here, we study the integrated scores over multiple tasks in Atari 2600 and DM Control from RL-Unplugged. Atari 2600 consists of 46 Atari games, and DM Control has 9 different tasks. We use the performance of 7 different baselines⁹ in our analysis.

⁹For our analysis we used the data from the ancillary files of [Schrittwieser et al., 2021], which can be found in <https://arxiv.org/src/2104.06294v1/anc>.

Table 2: Top 3 performing models on LRA depending on which subset of tasks we select.

Task	Best Model	Rank-2	Rank-3
t_1 (Text only)	Linear Transformers	Performer	Transformer
t_2 (Retrieval only)	Sparse Transformers	BigBird	Longformer
t_3 (ListOps only)	Reformer	Synthesizer	Transformer
t_4 (Image only)	Sparse Transformer	Performer	Transformer
t_5 (Path only)	Performer	Linformer	Linear Transformers
t_1+t_2	BigBird	Sparse Transformer	Transformer
t_1+t_3	Transformer	BigBird	Synthesizer
t_1+t_4	Linear Transformer	Performer	Transformer
t_1+t_5	Performer	BigBird	Transformer
t_2+t_3	BigBird	Transformer	Longformer
t_2+t_4	Sparse Transformer	BigBird	Transformer
t_2+t_5	BigBird	Sparse Transformer	Performer
t_3+t_5	Linformer	BigBird	Transformer
t_3+t_4	Transformer	Synthesizer	Longformer
t_4+t_5	Performer	Linear Transformer	Sparse Transformer
$t_1+t_2+t_3$	BigBird	Transformer	Synthesizer
$t_1+t_2+t_4$	Sparse Transformer	Transformer	BigBird
$t_1+t_2+t_5$	Performer	Linear Transformer	Transformer
$t_2+t_3+t_4$	Transformer	Longformer	Synthesizer
$t_2+t_3+t_5$	BigBird	Transformer	Longformer
$t_3+t_4+t_5$	BigBird	Transformer	Longformer
$t_1+t_2+t_3+t_4$	Transformer	BigBird	Longformer
$t_1+t_3+t_4+t_5$	BigBird	Transformer	Longformer
$t_1+t_2+t_4+t_5$	Sparse Transformer	Performer	BigBird
$t_2+t_3+t_4+t_5$	BigBird	Transformer	Longformer
$t_1+t_2+t_3+t_4+t_5$ (LRA Score)	BigBird	Transformer	Longformer

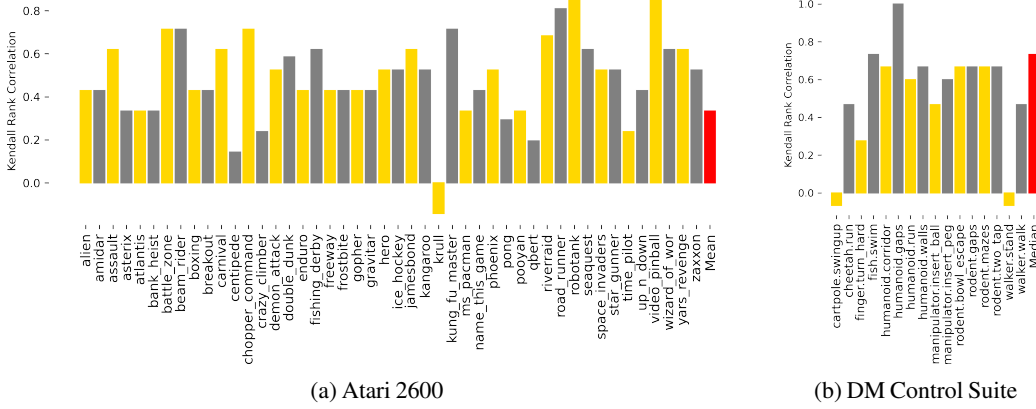


Figure 5: Rank correlation between the mean score and scores from each individual dataset.

Figure 5a presents the Kendall rank correlation when ranking different models based on their human normalized performance on each task vs the *median* human normalized performance across all tasks. We also show the correlation between median and mean human normalized performance on Atari. Although many papers reported mean performance on Atari as the integrated score, it is becoming a standard to report median since the mean is potentially less informative, as it is dominated by a few games (e.g. Atlantis) where agents achieve scores orders of magnitude higher than humans do. Figure 5b also shows the Kendall rank correlation of the mean performance across all tasks with performance on each task as well as the median. In both cases, it can be seen that the ranking of models based on individual tasks can widely disagree the ranking from the integrated score (average rank correlation in Figure 5a is ≈ 0.49 and in Figure 5b is ≈ 0.54), indicating how solely reporting the integrated score can send a potentially wrong signal for choosing the best model.

B Community bias example

The GLUE benchmark was pitched as a general language understanding benchmark and is an aggregation of 8 datasets that have been previously proposed [Williams et al., 2017]. We use this as an example of a community bias. To date, most pretrained language models (e.g., Transformers) evaluate on GLUE. This includes widely recognized and cited papers such as BERT [Devlin et al., 2018], ALBERT [Lan et al., 2019], RoBERTa [Liu et al., 2019], XLNet [Yang et al., 2020], ELECTRA and many others. To this date, almost all ($> 99\%$, taking into account the papers that we’ve potentially missed.) widely recognized (i.e., > 500 citations) pretrained LM paper evaluates on the GLUE benchmark. Given the popularity of GLUE for evaluating these models, it is only imperative that newly proposed models have to shine on GLUE in order in order to increase their likelihood of acceptance (in the literal sense or by the community).

Here, it is good to note that seven out of eight tasks in GLUE are actually *matching* tasks which model the relationship between two or more sequences. While it is still unclear how much this problem formulation has to do with natural language understanding, it is clear that this problem formulation favors a certain class of models (e.g., Transformers which has this baked-in cross attention in the encoder). It is easy to see that this conflates an actual advantage in problem formulation (and input setting) with the ability of an encoder model to learn textual representations. While one may argue that a method should reap rewards even for a problem formulation advantage, it is also good to note that many of these cross attention setups are infeasible in practice at scale [Guo et al., 2019, Seo et al., 2018]. It is also interesting that, if the tasks in GLUE were swapped for other equally plausible and practical tasks, we might actually encourage the development of alternative architectures such as pretrained ConvNets in NLP [Tay et al., 2021b].

C Rigging the lottery: additional case study

C.1 ALE and evaluation setup inconsistencies

An example of a benchmark that hundreds of papers have used it as a test bed, while simultaneously employing a number distinct experimental evaluation protocols is the Arcade Learning Environment (ALE) which is based on Atari 2600 games [Mnih et al., 2013]. The main aspects in which evaluation setups in different papers using ALE diverge are different metrics used for summarizing agent performance, and the different mechanisms used for injecting stochasticity in the environment [Machado et al., 2018].

For example, different assumptions can be made for determining episode termination. While in some publications episodes terminate when the game is over [Bellemare et al., 2013, Hausknecht et al., 2014, Liang et al., 2015, Lipovetzky et al., 2015, Martin et al., 2017], while others papers choose to terminate the training episodes for a subset of the games when the agent loses a life [Mnih et al., 2016, Nair et al., 2015, Wang et al., 2016, Van Hasselt et al., 2016].

Another major disagreement in evaluation strategies for ALE, also comes from using different parameters used for the evaluation setup. For example, some papers use a non-default value for the skipframe parameter¹⁰ in their baseline models [Mnih et al., 2015]. Alternatively in some publication, methods are evaluated for each 2×10^5 frames [Pritzel et al., 2017], while in others methods are evaluated every 10^6 frames [Mnih et al., 2013, 2016]. Another observation is the difference between the number of games used in the evaluation setups. For instance, Mnih et al. [2015] use 49 games, while Van Hasselt et al. [2016], Wang et al. [2016] use 57. Moreover, for the hyper-parameter tuning, sometimes papers use the entire suite of games as the validation set [Bellemare et al., 2013], while in other cases hyperparameters are optimized on a per-game basis [Jaderberg et al., 2016].

Yet another inconsistency is in reporting of the results in terms of the variety of different summary statistics used to describe them, which makes direct comparison between publications difficult [Machado et al., 2018]. To make matters worse sometimes sufficient statistics to make a judgment on the quality of the models are not provided. As an example, in [Bellemare et al., 2013], the main results are reported as the average performance of the method as well as the best run without mentioning the variance or the standard error of the mean. This is particularly problematic for reinforcement learning, where it has been shown that often the variance between runs can be so large as to create statistically different distributions just by varying random seeds [Henderson et al., 2018].

The final contentious aspect of ALE that we highlight is the way that various publications choose to inject stochasticity into the environment. ALE is fully deterministic, thus it is possible to get good scores by simply memorizing the “right” action sequence, rather than learning to make good decisions in a variety of game scenarios (i.e. learning an open-loop policy). With this in mind, to encourage and evaluate agent robustness, various ideas were developed to add forms of stochasticity to ALE [Bellemare et al., 2013]. Unfortunately, these methods are not necessarily consistent with each other.

¹⁰When predicting the action given the state, it is often done for every k -th frame, where k is the skipframe hyper-parameter.

D Popular public benchmarks for evaluating recommend systems

Table 3 presets the list of publicly available datasets for recommender systems used by the community for evaluation.

Table 3: List of popular offline datasets used for evaluating recommender systems.

Dataset	Number of examples	Users	Items	Sparsity
MovieLens 1M ¹¹	1,000,209	3706	6040	95.53%
MovieLens 20M ¹²	13,501,622	138,159	16,954	99.42%
Amazon Product Review (Movies & TV) ¹³	505K	22,147	178,086	99.98%
Amazon Product Review (Video Games)	46K	2,670	47,063	99.96%
Yahoo Movies ¹⁴	221,367	7,642	11,915	99.76%
Pinterest ¹⁵	1.5M	9916	55187	99.73%
Xing ¹⁶	1,450,300	65,347	20,778	99.89%
Taobao ¹⁷	100M	968K	4M	99.98%
Last.FM ¹⁸	42,346	1,872	3,846	99.41%
Book-Crossing ¹⁹	172,576	19,676	20,003	99.96%

¹¹<https://grouplens.org/datasets/movielens/1m/>

¹²<https://grouplens.org/datasets/movielens/20m/>

¹³<https://jmcauley.ucsd.edu/data/amazon/>

¹⁴<https://webscope.sandbox.yahoo.com/catalog.php?datatype=r>

¹⁵<https://paperswithcode.com/dataset/pinterest>

¹⁶<http://www.recsyschallenge.com/2017/>

¹⁷<https://tianchi.aliyun.com/dataset/dataDetail?dataId=649>

¹⁸<http://ocelma.net/MusicRecommendationDataset/>

¹⁹<http://www2.informatik.uni-freiburg.de/~chiegler/BX/>